GLUEING OF MORPHISMS NEAR A CLOSED POINT OF THE LIMIT

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Editorial note. No omissions, hints or 'immediate' steps in the counted span.

Allowed standard prerequisites (editorial).

- Glueing of schemes and morphisms. Schemes, glueing lemmas

Source reference. Original reference 01JA: sources/stacks-chow/tags.txt, source lines 2007–2007. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 01ZC: sources/stacks-chow/tags.txt, source lines 2587–2587. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01YX: sources/stacks-chow/tags.txt, source lines 2572–2572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let X and Y be schemes over S. Assume Y is locally of finite presentation over S. Let $x \in X$ be a closed point such that $U = X \setminus \{x\} \to X$ is quasi-compact. With $V = \mathrm{Spec}(\mathcal{O}_{X,x}) \setminus \{x\}$ there is a bijection*

$$\{\textit{morphisms } X \to Y \textit{ over } S\} \longrightarrow \left\{ \begin{array}{c} (a, b) \textit{ where } a : U \to Y \textit{ and } b : \mathrm{Spec}(\mathcal{O}_{X,x}) \to Y \\ \textit{are morphisms over } S \textit{ which agree over } V \end{array} \right\}$$

Proof. Let $W \subset X$ be an open neighbourhood of x. By glueing of schemes, see Schemes, Section 01JA the result holds if we consider pairs of morphisms $a : U \to Y$ and $c : W \to Y$ which agree over $U \cap W$. We have $\mathcal{O}_{X,x} = \mathrm{colim}\, \mathcal{O}_W(W)$ where W runs over the affine open neighbourhoods of x in X. Hence $\mathrm{Spec}(\mathcal{O}_{X,x}) = \lim W$ where W runs over the affine open neighbourhoods of s. Thus by Proposition 01ZC any morphism $b : \mathrm{Spec}(\mathcal{O}_{X,x}) \to Y$ over S comes from a morphism $c : W \to Y$ for some W as above (and c is unique up to further shrinking W). For every affine open $x \in W$ we see that $U \cap W$ is quasi-compact as $U \to X$ is quasi-compact. Hence $V = \lim W \cap U = \lim W \setminus \{x\}$ is a limit of quasi-compact and quasi-separated schemes (see Lemma 01YX). Thus if a and b agree over V, then after shrinking W we see that a and c agree over $U \cap W$ (by the same proposition). The lemma follows. □

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